

Quantum Measurement: Separating the Physical Burdens and Testing Outcome-Conditioned Continuation

The Measurement Problem Is Not One Question

A quantum measurement apparatus can answer one physical question while leaving another untouched. An interaction may be demonstrably sensitive to a chosen observable without leaving a durable record. A stable record may exist while the apparatus is miscalibrated with respect to the quantity it is said to measure. Two apparatuses can realize the same outcome statistics while disturbing the measured system differently. A collection of local records can be mutually compatible without establishing that exactly one outcome possesses theory-wide physical standing. Decoherence can stabilize a preferred pointer structure without, by that fact alone, settling the interpretation of the resulting branches or records. These are not semantic distinctions. They correspond to different experimental or theoretical jobs and, importantly, to different ways a measurement claim can fail.¹

Modern measurement theory already contains one especially clean example of this separation. A positive-operator-valued measure specifies the probabilities assigned to outcomes, but it does not in general determine the postmeasurement dynamics. Quantum instruments carry the additional information: they associate each classical outcome with a quantum operation describing the corresponding continuation of the measured system. Consequently, the same measurement statistics can be compatible with different postmeasurement states and different later sequential predictions.²

¹ Maximilian Schlosshauer, “Decoherence, the measurement problem, and interpretations of quantum mechanics,” *Reviews of Modern Physics* 76, 1267–1305 (2005), DOI 10.1103/RevModPhys.76.1267. The source is used here for the narrower point that decoherence, classical-record emergence, and the full foundational measurement problem should not be collapsed into one claim.

² The quantum-instrument formalism supplements outcome statistics with the outcome-conditioned quantum operation. The distinction is standard but crucial here: a POVM fixes probabilities while generally underdetermining the postmeasurement state and therefore later sequential statistics. See, for example, N. Saberian, S. J. Akhtarshenas, and F. Shahbeigi, “Measurement sharpness and disturbance tradeoff,” *Physical Review A* 109, 012201 (2024), DOI 10.1103/PhysRevA.109.012201. For the corresponding operational perspective, see Roman Stricker et al., “Characterizing Quantum Instruments: From Nondemolition Measurements to Quantum Error Correction,” *PRX Quantum* 3, 030318 (2022), DOI 10.1103/PRXQuantum.3.030318.

Treating “the measurement” as exhausted by its POVM therefore discards physical information that becomes observable as soon as another interaction follows.

The same caution applies to decoherence. Environment-induced decoherence can suppress interference between preferred pointer states and explain why particular apparatus observables acquire robust, effectively classical records. That achievement is substantial, but it does not license the substitution of one measurement burden for every other one. Reviews of decoherence and quantum measurement have long emphasized both the dynamical importance of environment-induced superselection and the continuing distinction between decoherence and the broader interpretive measurement problem.³ Whether a pointer basis becomes dynamically stable, whether a physical record is available, whether only one outcome is ultimately taken to stand, and what quantitative rule assigns outcome weights are different questions.

This work therefore begins with a deliberately severe criterion for separation. A proposed measurement burden is treated as independent only if three things can be stated without borrowing the answer from another burden: what exact job it performs, what the weakest physical structure capable of performing that job looks like, and what specific capacity disappears when that structure fails while neighboring measurement structure survives. If no such separating state exists, the distinction is decorative. If one does exist, collapsing the two questions together hides a physical assumption.

Applied across quantum measurement, that test yields a dependency structure rather than a single measurement “step.” Before anything else, a measurement claim needs an admitted physical seat: a source or system, an apparatus or probe, a declared event structure, a coupling or instrument, and enough calibration to justify saying that the apparatus measures the target it is claimed to measure. A completed interaction may optionally be subjected to a controlled-reversion test, but reversibility is not constitutive of measurement. Once a measurement seat is admitted, one may separately ask whether an outcome has become an incorporated physical record; whether that record remains available to condition later events; whether physically different carriers encode the same binding content; whether those local descriptions

³ W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical,” *Reviews of Modern Physics* 75, 715–775 (2003), DOI 10.1103/RevModPhys.75.715; Schlosshauer, *Reviews of Modern Physics* 76, 1267–1305 (2005). Zurek provides the pointer-state and environment-induced-superselection framework used here; Schlosshauer is useful for the boundary between that dynamical account and stronger measurement-problem claims.

can be held together as one physical account; whether exactly one outcome class has the realized standing claimed by a theory; what quantitative measure weights the admitted outcomes; and what continuation rule governs later prediction or interaction after a particular outcome.

The accompanying audit architecture compresses these jobs as M0 and A–H. Those labels are bookkeeping, not the argument. What matters is that the jobs discriminate. A calibrated measurement need not leave a durable record. A durable record can be miscalibrated. Several records can agree without establishing unique global outcome standing. A theory can supply unique outcome standing without thereby deriving the Born measure. And one-shot outcome probabilities can remain fixed while different postmeasurement dynamics generate different predictions for the next experiment.

Several of these questions remain interpretation-sensitive or experimentally difficult. That is not a defect in the decomposition. The purpose of separating them is precisely to prevent an unresolved answer at one level from being smuggled into another. A globally unitary description, for example, can preserve one coherent dynamical whole while containing multiple decohered outcome sectors. The existence of that whole therefore does not by itself establish exactly-one-outcome standing. Conversely, theories with sharply different accounts of what is physically actual can reproduce the same ordinary Born statistics. Outcome standing therefore does not itself determine probability. Probability, in turn, does not determine what physical or predictive continuation follows a registered outcome.

The last distinction is unusually useful experimentally. A laboratory need not settle whether the quantum state is ontic, whether branching worlds are fundamental, or whether an outcome is absolute in order to ask whether two measurement procedures with the same one-shot statistics produce different later statistics. Measurement continuation can be treated operationally first. A theory may subsequently interpret the continuation as an ontic state change, a relational state, an effective conditional state, an information update, or another object. What matters for the experiment is whether the declared rule makes definite predictions for what comes next.

That observation changes the form of the measurement question. Instead of asking immediately why a single outcome occurs, or what “collapse” really is, one can ask a narrower question that standard quantum measurement theory already knows how to make precise:

Once the preparation and the first measurement have been independently fixed, what information is allowed to survive into the statistics of a later measurement?

The experiment developed here begins from that question. Before it can be asked cleanly, however, the first measurement itself has to be made physically honest.

What a Lawful Measurement Claim Has to Hold Fixed

Calling an interaction a measurement does not make it one. A device can respond reproducibly to the wrong variable, attach the wrong label to a correct response, resolve a finer distinction than the analysis assumes, or stabilize a pointer structure that does not correspond to the target observable as advertised. Quantum calibration is therefore an empirical task in its own right. An unknown measurement apparatus can, in principle, be characterized through independently warranted preparations and tomographic structure rather than by taking its nominal design specification as its measurement identity.⁴

For the present experiment, a lawful measurement claim requires enough physical information to identify what is being measured, by what apparatus, through what event structure, under what interaction, and with what calibration or faithfulness relation between the target content and the registered outcomes. None of this requires a universal representation of measurement. A nonrelativistic subsystem model may naturally use Hilbert spaces, effects, and instruments. A field-theoretic or detector-based setting may require local algebras, probes, channels, or another domain-specific representation. The burden is to establish the physical identity of the measurement at the scope being claimed, not to privilege one mathematical notation.

This admission step prevents an important circularity. The later existence of a stable record cannot be the sole reason for declaring what observable was measured. A robust record may establish that some physical distinction became stable while leaving open whether the apparatus tracked the intended source variable. Likewise, agreement with a familiar Born-rule prediction may legitimately help calibrate an apparatus under standard quantum theory, but that same agreement cannot simultaneously function as

⁴ G. M. D'Ariano, L. Maccone, and P. Lo Presti, "Quantum Calibration of Measurement Instrumentation," *Physical Review Letters* 93, 250407 (2004), DOI 10.1103/PhysRevLett.93.250407. The relevance here is methodological: measurement identity and calibration can themselves be experimentally characterized rather than assumed from apparatus design intent.

independent evidence in an argument that claims to derive the Born rule itself. Calibration and derivation are different inferential routes. QCCT takes the empirical-conformance route: ordinary quantum mechanics may be used to characterize the apparatus because the proposed experiment is not a derivation of quantum probability.

Pointer stability requires the same discipline. Decoherence and environment-induced superselection provide physical mechanisms through which some apparatus observables become more robust than others; environmental interaction can suppress interference between alternative pointer states and thereby privilege a stable readout structure.⁵ But that preferred readout structure is a feature of the physical interaction architecture. It cannot simply be selected after the fact because a particular mathematical basis happens to be convenient for the intended interpretation. If several outcome decompositions are mathematically available, the experiment must warrant why the declared event structure is the one physically implemented.

Once that measurement seat is fixed, standard quantum theory distinguishes two further objects that are easily conflated. One-shot outcome statistics can be represented by effects E_i satisfying

$$p(i \mid \varrho) = \text{Tr}(E_i \varrho).$$

A quantum instrument adds outcome-conditioned operations $\mathcal{I}_i$ such that

$$p(i \mid \varrho) = \text{Tr}[\mathcal{I}_i(\varrho)].$$

If a later measurement has effects F , the experimentally relevant sequential probability becomes

$$p(i, j \mid \varrho) = \text{Tr}[F \mathcal{I}_i(\varrho)].$$

The last equation exposes the measurement burden that the proposed experiment targets. The same effects E_i —and therefore the same one-shot outcome statistics—can be associated with different admissible instruments $\mathcal{I}_i$. Once a later probe is introduced, those instruments can yield different values of $p(i, j \mid \varrho)$. The

⁵ Zurek, *Reviews of Modern Physics* 75, 715–775 (2003). The limited claim used here is that physical environmental coupling can privilege robust pointer structure. QCCT does not infer unique outcome standing, a probability rule, or a universal state-update law from pointer stability.

outcome statistics alone therefore do not specify the complete physical or predictive continuation of a measurement.⁶

This distinction matters because an ordinary quantum measurement can be perfectly well calibrated and still fail to follow the particular postmeasurement rule an analyst initially expected. A real apparatus need not implement the ideal Lüders operation merely because its outcome projectors have been correctly identified. The instrument may contain ordinary disturbance, leakage, dissipation, additional unitary evolution, or other device-specific effects. None of those constitutes foundationally anomalous physics. They are reasons to identify the actual instrument before using sequential measurements to make stronger claims.

QCCT therefore does not begin by asking whether the projection postulate is universally “true.” That question is both too broad and experimentally unnecessary. Fiorentino and Weigert recently developed a family of generalized state-update theories that departs from the usual Lüders continuation while preserving substantial consistency conditions at the scopes they study.⁷ Their framework is useful here because it makes the continuation burden explicit and supplies concrete comparison theories. But those theories do not receive evidentiary priority over ordinary apparatus physics. A different standard instrument, preparation leakage, temporal memory, an incomplete system boundary, or another independently warranted ordinary mechanism must be allowed to account for the sequential data first.

That ordering is central to the proposal. A deviation from an ideal Lüders prediction is not, by itself, evidence for generalized quantum dynamics. A failure of one fitted instrument is not, by itself, evidence for generalized quantum dynamics. Even a preparation-dependent sequential signal is not interpretable until the experiment has asked whether the preparation procedure left physically accessible information outside the declared quantum state. The target can be blamed only after the serious ordinary

⁶ The formal distinction between effects and instruments is load-bearing for QCCT. An instrument carries the classical outcome together with the associated quantum operation, so sequential experiments can distinguish measurement implementations that one-shot outcome statistics cannot. See Roman Stricker et al., “Characterizing Quantum Instruments: From Nondemolition Measurements to Quantum Error Correction,” *PRX Quantum* 3, 030318 (2022), DOI 10.1103/PRXQuantum.3.030318. The artifact relies only on the standard consequence that identical POVM effects need not uniquely specify the postmeasurement operation.

⁷ V. Fiorentino and S. Weigert, “Beyond the projection postulate and back: Quantum theories with generalized state-update rules,” *Physical Review A* 113, 012204 (2026), DOI 10.1103/2zpm-jsh7. Their framework supplies explicit generalized state-update rules and operational conditions distinguishing them from Lüders updating. QCCT uses these rules as named theory-side foils; it does not treat their existence as empirical evidence against standard quantum mechanics.

routes capable of producing the same observation have been admitted and tested at the claimed scope.

The physical question can now be narrowed substantially. We need a preparation whose standard quantum description is fixed, a first measurement whose identity is independently admitted, and a later probe capable of revealing whether the continuation carries information that a standard quantum instrument is not allowed to retain.

The next step is to show that such a distinction is not merely hypothetical. A specific pair of preparation procedures can be represented by the same density operator in standard quantum mechanics while a published generalized-update rule predicts that their postmeasurement continuations differ. That separation is the mathematical center of QCCT.

After the Outcome: The Continuation Burden

Once a measurement has been admitted and its one-shot outcome statistics characterized, a further question remains: what description governs the system, output, or predictive state from which later interactions proceed? In the ideal projective case this question is often answered almost automatically by invoking the projection postulate. For an outcome represented by P_i , the familiar Lüders rule selects the corresponding projected state. But that familiar case can obscure the more general physical burden. The experimental question is not whether one preferred textbook formula can be written after an outcome. It is whether the rule used after that outcome is independently warranted by what the apparatus actually does.

Quantum instruments make the separation operational: two implementations can agree on the available outcomes and their one-shot probabilities while producing different conditional dynamics, distinguishable only when another interaction follows.

That observation also prevents an unnecessary interpretive commitment.

“Outcome-conditioned continuation” is deliberately broader than “physical collapse.” A theory may understand the postmeasurement object as an ontic state of the system, a relational state, an effective conditional state, an operational predictive state, an epistemic update, or another theory-specific object. QCCT does not decide among

those readings in advance. It asks only whether the governing theory supplies a continuation rule and whether that rule correctly predicts the later experiment.

Nor does this question require the Lüders rule to be treated as the only physically respectable alternative. Measurements associated with the same one-shot effects can be implemented by different ordinary quantum instruments. A non-Lüders postmeasurement state can therefore be entirely standard quantum mechanics. Indeed, Lüders and von Neumann updating for degenerate measurements have already been experimentally discriminated using NMR, with the tested apparatus favoring Lüders behavior.⁸ QCCT consequently makes no claim to be the first experiment capable of distinguishing state-update rules, nor does a departure from an ideal Lüders prediction constitute its foundational target.

The more useful question is whether the ordinary quantum-instrument framework itself imposes a continuation invariance that a well-defined alternative theory can violate. Fiorentino and Weigert recently approached the projection postulate from exactly the complementary direction: instead of assuming Lüders updating and asking how to derive it, they constructed explicit generalized state-update rules subject to consistency requirements, including unique postmeasurement descriptions and restrictions associated with signaling and composite systems.⁹ Their framework produces genuine foil theories rather than arbitrary nonlinear functions chosen to fit a prospective anomaly. It also isolates operational properties by which the Lüders rule can be distinguished from those alternatives.

One of those properties is preparation indistinguishability. Standard quantum mechanics represents a statistical preparation by its density operator. Once two preparation procedures are represented by the same density operator, a fixed quantum instrument acts on the same mathematical input in both cases. It cannot condition its output on which ensemble decomposition happened to generate that density operator. The preparation history has disappeared from the standard state description.

⁸ C. S. Sudheer Kumar, A. Shukla, and T. S. Mahesh, “Discriminating between Lüders and von Neumann measuring devices: An NMR investigation,” *Physics Letters A* 380, 3612–3616 (2016), DOI 10.1016/j.physleta.2016.09.004. The experiment implemented a four-qubit NMR protocol designed to discriminate the two state-update rules for a degenerate measurement and reported support for Lüders behavior at the tested device. Its relevance here is chiefly negative: QCCT should not be presented as the first experimental discrimination of measurement-update rules.

⁹ V. Fiorentino and S. Weigert, “Beyond the projection postulate and back: Quantum theories with generalized state-update rules,” *Physical Review A* 113, 012204 (2026), DOI 10.1103/2zpm-jsh7. The paper constructs explicit unconventional state-update rules subject to consistency requirements, including unique postmeasurement descriptions and constraints associated with signaling and composite systems, and identifies operational properties that distinguish Lüders updating from the resulting foils.

A generalized nonlinear update rule need not share that property. Fiorentino and Weigert distinguish a proper mixture—classical uncertainty over which preparation was actually performed—from an improper mixture arising, for example, as the reduced state of a larger entangled system. For a proper mixture, or Gemenge, their generalized framework can update each physically prepared component separately before the ensemble is averaged. When the update map is nonlinear, two different Gemenge associated with the same ordinary density operator can therefore produce different postmeasurement descriptions.¹⁰

That distinction identifies a much cleaner experimental target than the question “Does Lüders hold?” The target is not one ideal projection rule versus another ordinary instrument. It is the stronger invariance implied by the standard instrument formalism:

If two real preparation procedures supply the same admitted density operator to the same fixed quantum instrument, can their preparation decomposition affect the statistics of a later measurement?

Under standard quantum instruments, the answer is no. Under explicit generalized-update foils, the answer can be yes.

The passive-measurement theory developed earlier by Fiorentino and Weigert makes the physical contrast especially transparent. In that foil, measurement outcomes still occur with Born probabilities, but measurement does not collapse the quantum state. The resulting theory preserves a surprising amount of familiar quantum structure while producing different predictions in protocols that depend on postmeasurement states.¹¹ Their later generalized framework places that passive theory at one endpoint of a continuous family of state-update rules. QCCT uses that family not because it is presumed correct, but because it gives the experiment something unusually valuable: a named, internally specified alternative with an exact prediction rather than an unspecified category called “nonstandard quantum mechanics.”

¹⁰ Ibid.

¹¹ V. Fiorentino and S. Weigert, “A quantum theory with non-collapsing measurements,” *Physics Letters A* 559, 130903 (2025), DOI 10.1016/j.physleta.2025.130903. This earlier foil retains Born-distributed measurement outcomes while making the measurement passive with respect to state update. It is used here only to give physical intuition for the $\lambda = 1$ endpoint of the later generalized family.

The experiment can therefore be designed asymmetrically in favor of ordinary physics. Standard quantum mechanics receives the broader comparison class: any admissible fixed quantum instrument, and later, if necessary, a finite-horizon process description capable of carrying ordinary apparatus memory. The generalized theory receives no such freedom. It must enter through a predeclared update family with a predeclared observable difference. Only after the ordinary comparison class has had that first refusal does the foil become evidentially interesting.

The simplest realization requires only a qubit.

QCCT-I: Same Density Operator, Different Preparation Gemenge

Consider two physically implemented randomized preparation procedures. The first, G_Z , prepares $|0\rangle$ or $|1\rangle$ with equal probability:

$$G_Z = \{ (1/2, |0\rangle\langle 0|), (1/2, |1\rangle\langle 1|) \}.$$

The second, G_X , prepares $|+\rangle$ or $|-\rangle$ with equal probability:

$$G_X = \{ (1/2, |+\rangle\langle +|), (1/2, |-\rangle\langle -|) \},$$

where

$$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

and

$$|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}.$$

In standard quantum mechanics the two procedures have the same density operator:

$$\varrho_Z = \varrho_X = \frac{1}{2}I.$$

This equality is the center of the experiment, but it has to be read correctly. G_Z and G_X are not merely two algebraic decompositions that an analyst chooses to write for $\frac{1}{2}I$. They are two real preparation procedures: in one case a classical randomizer selects between the Z-basis states; in the other it selects between the X-basis states. Standard quantum theory represents both procedures by the same density operator

once the classical preparation label is discarded. The generalized foil retains their Gemenge structure when defining the nonlinear state update.¹²

Now let the first measurement contain the rank-one outcome

$$P_0 = |0\rangle\langle 0|$$

with complementary outcome

$$P_1 = |1\rangle\langle 1|.$$

For either preparation procedure, the one-shot Born probability of obtaining P_0 is

$$\Pr(P_0 \mid G_Z) = \Pr(P_0 \mid G_X) = 1/2.$$

The two experimental branches therefore agree before the continuation question is entered. They supply the same standard density operator and the same probability for the first outcome.

Suppose first that ordinary quantum measurement theory is correct at the declared experimental seat. Let $\mathcal{J}_0$ denote the quantum operation associated with outcome P_0 . A standard quantum operation is linear, completely positive, and trace non-increasing.¹³ Because its two inputs are equal,

$$\mathcal{J}_0(\rho_Z) = \mathcal{J}_0(\rho_X).$$

For any later measurement effect F ,

$$\text{Tr}[F\mathcal{J}_0(\rho_Z)] = \text{Tr}[F\mathcal{J}_0(\rho_X)].$$

Equivalently, define the joint preparation-indistinguishability witness

$$\Delta_PI(F) = \Pr(P_0 \text{ then } F \mid G_X) - \Pr(P_0 \text{ then } F \mid G_Z).$$

¹² Fiorentino and Weigert, *Physical Review A* 113, 012204 (2026). Their proper-mixture discussion is load-bearing for QCCT. For a Gemenge $G = \{ (p_i, \rho_i) \}$, the generalized rule acts on the individual component states and then forms the weighted postmeasurement mixture. They explicitly distinguish this from the treatment of pure states and improper mixtures. The same paper gives the standard quantum-operation premise used for the null: quantum operations are linear, completely positive, and trace non-increasing. Its passive-rule example also uses the two equal-weight Z- and X-basis preparation Gemenge associated with the same maximally mixed density operator and shows that their postmeasurement descriptions can differ.

¹³ Ibid.

For any fixed standard quantum instrument acting on the admitted state at the declared reference,

$$\Delta_{\text{PI}}(F) = 0.$$

This is the important null. It does not assume Lüders updating. It does not require the first measurement to be minimally disturbing. It does not require a particular postmeasurement density matrix. It follows from the more general fact that the same linear operation acts on the same density operator. A non-Lüders ordinary instrument may change both branches substantially; it still cannot distinguish which ensemble decomposition generated $\frac{1}{2}I$ unless some additional physical information about that preparation reaches the instrument.

The generalized foil is different because its state-update prescription for a proper mixture is not obtained by applying one nonlinear function directly to the averaged density operator. If

$$G = \{ (p_i, \varrho_i) \},$$

then the proper-mixture rule applies the update to each component and only afterward forms the weighted result:

$$\Omega_{G,\lambda}(P_x) = \sum_i p_i \omega_\lambda(P_x, \varrho_i).$$

This componentwise step is load-bearing. If one instead replaced G_Z and G_X by the common matrix $\frac{1}{2}I$ before applying the nonlinear rule, the preparation distinction being tested would already have been erased. Fiorentino and Weigert make the proper/improper-mixture distinction explicit for precisely this reason.¹⁴

For the single-system probability-amplifying family used here, define

$$G\lambda(P_x) = (1 - \lambda)P_x + \lambda I,$$

with $0 \leq \lambda \leq 1$, and

$$\omega_\lambda(P_x, \varrho) = \text{Tr}(P_x \varrho) \cdot [\sqrt{G\lambda(P_x)} \varrho \sqrt{G\lambda(P_x)} / \text{Tr}(G\lambda(P_x) \varrho)].$$

¹⁴ Ibid.

At $\lambda = 0$ the family returns the Lüders rule. At $\lambda = 1$ it reaches the passive, noncollapsing endpoint. For nonzero λ the rule is nonlinear in the relevant sense and can violate preparation indistinguishability.¹⁵

Apply the family to outcome P_0 . For G_Z , only the $|0\rangle$ component contributes to that outcome, giving the subnormalized continuation

$$\Omega_{Z,\lambda}(P_0) = \frac{1}{2}P_0.$$

Its trace is $\frac{1}{2}$, as required by the unchanged one-shot probability. Conditioned on obtaining P_0 , the continuation is simply

$$\varrho_{Z|0,\lambda} = P_0.$$

The G_X preparation behaves differently. Updating $|+\rangle$ and $|-\rangle$ separately produces equal diagonal contributions and opposite off-diagonal contributions. The off-diagonal terms cancel when the two proper-mixture components are averaged, leaving

$$\Omega_{X,\lambda}(P_0) = [P_0 + \lambda P_1] / [2(1 + \lambda)].$$

Its trace is also $\frac{1}{2}$. The first measurement therefore still cannot distinguish the two preparations by outcome frequency alone. Conditioned on P_0 , however,

$$\varrho_{X|0,\lambda} = [P_0 + \lambda P_1] / (1 + \lambda).$$

A later measurement of P_1 now separates the theories. For G_Z ,

$$\Pr(P_1 \text{ later} \mid P_0, G_Z) = 0,$$

whereas for G_X the generalized family predicts

$$\Pr(P_1 \text{ later} \mid P_0, G_X) = \lambda / (1 + \lambda).$$

Define the conditional QCCT witness

¹⁵ Fiorentino and Weigert, *Physical Review A* 113, 012204 (2026). The probability-amplifying family used in QCCT is one of their explicit generalized-update constructions. The QCCT-I derivation in this artifact reproduces the preparation-dependent witness from that published rule, and the $\lambda = 1$ endpoint can be checked directly against the paper's passive example. The paper identifies $\lambda = 0$ with the Lüders limit and $\lambda = 1$ with the passive limit; the family supplies the named foil rather than an ad hoc nonlinear function fitted after an observation.

$$W_{PI}(\lambda) = \Pr(P_1 \text{ later} \mid P_0, G_X) - \Pr(P_1 \text{ later} \mid P_0, G_Z).$$

The generalized family therefore predicts

$$W_{PI}(\lambda) = \lambda / (1 + \lambda).$$

If the experiment is analyzed using unconditioned joint counts instead, the first outcome occurs with probability $1/2$ in both branches, so

$$\Delta_{\text{joint}}(\lambda) = \lambda / [2(1 + \lambda)].$$

The endpoints provide an immediate check. At $\lambda = 0$,

$$W_{PI}(0) = 0$$

and

$$\Delta_{\text{joint}}(0) = 0,$$

recovering the preparation-indistinguishability prediction of standard Lüders updating. At the passive endpoint $\lambda = 1$,

$$W_{PI}(1) = 1/2$$

and

$$\Delta_{\text{joint}}(1) = 1/4.$$

The $\lambda = 1$ limit agrees with the proper-mixture counterexample given by Fiorentino and Weigert: two preparation Gemenge associated with the same maximally mixed density operator acquire different postmeasurement descriptions under the passive rule.¹⁶

The resulting experimental contrast is unusually sharp. The first outcome has the same standard probability in both preparation branches. The standard fixed-instrument class predicts no later dependence on the preparation decomposition. The selected generalized family predicts a continuous, analytic deviation from that null. A null experiment can therefore place a quantitative bound

¹⁶ Fiorentino and Weigert, *Physical Review A* 113, 012204 (2026).

on λ within this named family; a nonzero preparation-dependent signal can be compared directly with its predicted λ dependence.

Neither result carries the stronger foundational conclusions that are sometimes attached to measurement experiments. A null result would constrain this generalized-update family only under the admitted preparation, measurement, side-channel, and comparison assumptions. It would not prove the Lüders rule universally or exclude every nonlinear state-update theory. Conversely, observing $W_{PI} \neq 0$ would not by itself show that standard quantum mechanics has failed. The standard null applies only if the two branches actually present the same admitted quantum input to the same physical instrument. If the preparation procedure leaves accessible information in a resonator, higher level, control line, timing channel, environment, or other degree of freedom outside the declared qubit state, an ordinary apparatus can respond differently without violating the quantum-instrument formalism.

That is why QCCT is not complete when the witness is written down. The witness identifies the target. The next burden is to give ordinary quantum physics every serious opportunity to explain an apparent violation before the generalized foil receives any credit.

Giving Ordinary Quantum Physics First Refusal

The preparation-indistinguishability witness is only useful if its null is given a fair physical interpretation. The equation $\Delta_{PI}(F) = 0$ follows from a fixed standard quantum instrument acting on the same admitted density operator. A real experiment, however, is not handed an abstract density matrix and an abstract completely positive map. It is built from pulses, resonators, control electronics, higher energy levels, environmental modes, clocks, feed-forward paths, and apparatus states that may remember how an input was prepared. If any of those degrees of freedom carries information distinguishing G_Z from G_X , the two branches no longer present the same physical input to the same registered instrument. A preparation-dependent signal may then be entirely ordinary quantum physics.

QCCT therefore places the burden of ordinary explanation before the burden of generalized-update discrimination. The first question after any nonzero witness is not

whether the state-update law has failed. It is whether the experiment actually realized the reference under which the standard null was claimed.

That reference has to be operational rather than metaphysical. The statement $\rho_Z = \rho_X = \frac{1}{2}I$ means that the two preparations are represented by the same qubit state relative to a declared set of measurements or characterization procedures. No finite experiment establishes that the two physical preparation histories are identical in every possible respect. The relevant claim is narrower: within an independently selected and sufficiently rich premeasurement reference, the procedures are indistinguishable to the tolerance required by the experiment. Recent superconducting-qubit work has demonstrated that state spaces, effects, operational equivalences, and even process behavior can be reconstructed under substantially weaker assumptions than a fully trusted quantum model, while making the required tomographic-completeness assumption explicit.¹⁷ QCCT does not require that particular methodology, but it requires the same intellectual discipline: the reference by which the preparations are declared equivalent must be stated and must not be manufactured from the continuation result being tested.

Experimental imperfections make exact equivalence especially important. Real preparation procedures rarely land exactly on the ideal states specified on paper. This need not end the test. Earlier work on operational equivalence showed how experimentally realized “primary” preparations can be combined probabilistically to construct “secondary” preparations satisfying the required equivalence in the admitted operational representation, rather than treating near-equivalence as exact by fiat.¹⁸ The same strategy can be adapted to QCCT if the raw G_Z and G_X procedures do not meet the predeclared equivalence tolerance. What matters is that the mixture weights or correction procedure be determined independently of the later QCCT witness. Otherwise the target result would participate in constructing its own null.

¹⁷ A. Aloy et al., “Theory-independent monitoring of the decoherence of a superconducting qubit with generalized contextuality,” *Nature Communications* 17, 2474 (2026), DOI 10.1038/s41467-026-69030-x. The experiment develops a theory-independent prepare-and-measure reconstruction of a superconducting qubit in a generalized probabilistic framework and makes operational indistinguishability and the required tomographic-completeness assumption explicit. QCCT uses this work as methodological precedent for defining an operational reference, not as an execution of the continuation experiment.

¹⁸ M. D. Mazurek, M. F. Pusey, R. Kunjwal, K. J. Resch, and R. W. Spekkens, “An experimental test of noncontextuality without unphysical idealizations,” *Nature Communications* 7, 11780 (2016), DOI 10.1038/ncomms11780. Their primary laboratory procedures did not satisfy the desired operational equivalences exactly. The authors instead constructed secondary preparations and measurements as probabilistic mixtures of realized primary procedures so that the required equivalences held in the admitted operational representation. This is the specific precedent for QCCT's proposed secondary-preparation repair.

Even a successful premeasurement equivalence test is not enough if the preparation label can reach the measurement device through a channel that the declared qubit state does not contain. A Z-basis preparation pulse and an X-basis preparation pulse can leave different residual microwave fields. They can populate higher transmon levels differently, leave different resonator occupations, impose different idle times, heat the device differently, or place a classical controller in distinguishable internal states. A nominally random choice between ensemble components can itself be routed differently through control electronics. None of these possibilities is exotic. If the first measurement apparatus can access such a distinction, a later difference between G_Z and G_X does not violate the fixed-instrument prediction because the actual instrument input was never just $\frac{1}{2}I$.

The experiment should therefore treat preparation side channels as a positive search problem rather than as a disclaimer. Sequence duration can be matched. Preparation order can be randomized and interleaved. Resonators can be allowed to deplete. Leakage and thermal populations can be measured. Raw apparatus observables can be tested for residual predictability of the preparation label after conditioning on the admitted qubit state. Timing and calibration drift can be monitored. The exact controls belong to the laboratory implementing QCCT, but the logical rule is fixed in advance: if a preparation label remains physically available to the apparatus at the claimed scope, the preparation-indistinguishability test has not yet been entered.

Suppose those entry burdens close and a nonzero sequential difference remains. The next ordinary explanation is that the first measurement is not the idealized instrument initially used in the prediction. That possibility is handled by abandoning the ideal instrument before abandoning standard quantum mechanics. The comparison class should contain the most general fixed standard quantum instrument admitted by the physical reference, not merely the Lüders operation or a favored device model.

For a finite-dimensional system, each outcome operation can be represented through its Choi matrix and constrained to be completely positive and trace non-increasing, with the appropriate one-shot effect and loss structure. The instrument can be characterized on one set of preparations and later probes and tested on conditions not used to fit it. The purpose of held-out conditions is simple: an instrument model that merely reproduces the data from which it was reconstructed has not yet demonstrated predictive closure. QCCT needs the fitted ordinary instrument to predict later

statistics for unused preparations, probe bases, mixture weights, timing blocks, or other predeclared conditions.

If such an instrument predicts both preparation branches, the result is not a failed foundations experiment. It is **standard-instrument closure at the tested scope**. If the ideal Lüders model fails but another ordinary instrument succeeds, the correct result is even more specific: the apparatus implements a non-Lüders standard instrument. That is already scientifically useful. It is also exactly why QCCT does not equate “non-Lüders” with “nonstandard.”

A fixed instrument can still be too small a model if the apparatus retains memory. The physical transformation applied on one trial or at one time may depend on an environment that carries information from an earlier operation. Modeling such possibilities one mechanism at a time would create an obvious escape problem: cavity memory could be added after relaxation, another hidden mode after cavity memory, and another drift parameter after that. A comparison class that can expand indefinitely after seeing the residual is not a meaningful test.

The process-tensor framework supplies a cleaner ordinary-quantum boundary. Rather than requiring the experimenter to guess a particular memory mechanism, a finite-horizon process tensor represents the multi-time quantum process linking a declared sequence of interventions while allowing temporal correlations and non-Markovian memory within that registered process. In an early superconducting-qubit demonstration, the available control set did not span the complete space of quantum operations, so White and collaborators reconstructed a **restricted process tensor** over the linear span of the available unitary controls. For a single qubit that restricted control space had dimension ten, and the authors used a larger set of random unitaries to reconstruct and test its predictive behavior.¹⁹ The limitation was stated rather than hidden: the experiment characterized the process over the intervention space it could actually implement.

¹⁹ G. A. L. White et al., “Demonstration of non-Markovian process characterisation and control on a quantum processor,” *Nature Communications* 11, 6301 (2020), DOI 10.1038/s41467-020-20113-3. Because a complete basis of quantum operations was not available on the superconducting hardware used in that study, the authors reconstructed a restricted process tensor over the span of unitary controls. For the single-qubit case the resulting control space had dimension ten; the experiment used 28 random unitaries and tested process-tensor predictions beyond a minimal reconstruction basis. QCCT uses this as precedent for explicitly labeled restricted-process closure when a full intervention basis is unavailable.

More recently, full multi-time process tomography has been demonstrated on superconducting qubit hardware using sequential measure-and-prepare operations together with post-processing. Giarmatzi and collaborators report full characterization of a multi-time process, including correlated and specifically quantum non-Markovian noise, rather than only a Markovian channel sequence or restricted unitary-control slice.²⁰ Their result matters for QCCT because it removes a possible conceptual escape hatch. Finite-horizon process characterization on superconducting hardware is not merely a theoretical ideal; the relevant experimental machinery exists.

This does not mean that every QCCT implementation must perform full process-tensor tomography. If a fixed instrument already predicts the held-out data, the more expensive comparison may be unnecessary. If the laboratory can realize only a restricted intervention span, a restricted process tensor may be the strongest honest comparator. The distinction must simply be named. “Process-tensor closure” cannot mean full closure if only a subspace of interventions was tested.

The comparison order is therefore deliberately conservative. The experiment first asks whether a fixed standard instrument accounts for the data. If it does not, a finite-horizon process model is admitted where temporal memory is relevant. If that registered process model also fails, the experiment asks whether the physical system boundary was too narrow before it credits a generalized state-update rule.

Reference expansion, however, cannot become another unlimited escape. A resonator may be added to the modeled system if measurements show that it retains relevant information. A higher transmon level may be added if leakage is observed. A spectator qubit or controller state may be admitted if there is independent evidence that it couples to the sequence. What is not allowed is an indefinitely flexible “environment” introduced only because the registered model failed. An enlarged reference is another physical hypothesis and therefore requires its own warrant.

This produces a hierarchy of increasingly strong results without a jump from anomaly to ontology. A preparation-dependent signal may disappear when a side channel is closed, be captured by another standard instrument, or require a finite-horizon

²⁰ C. Giarmatzi, T. Jones, A. Gilchrist, P. Pakkiam, A. Fedorov, and F. Costa, “Multi-time quantum process tomography on a superconducting qubit,” *Quantum* 9, 1952 (2025), DOI 10.22331/q-2025-12-18-1952. The authors report full characterization of a multi-time process on superconducting hardware using sequential measure-and-prepare operations and post-processing, detecting both general temporal correlations and quantum-correlated noise. The result supplies direct feasibility support for QCCT’s finite-horizon process comparator at qubit scale.

process tensor or independently warranted reference expansion. Only after those routes have been discharged at the declared scope does the residual become a continuation problem for the registered ordinary comparison class.

Even then, the conclusion remains local. QCCT can establish that a registered ordinary quantum comparator is insufficient at a declared reference, horizon, and intervention surface; it cannot establish that every possible larger quantum description has failed. The generalized foil must meet the same discipline: its parameters and observable prediction are fixed in advance, and it earns only the claim supported by the residual.

The asymmetry is intentional. Ordinary quantum physics receives multiple opportunities to absorb the result because those opportunities correspond to known physical ways a laboratory can carry preparation history. The generalized-update family does not receive arbitrary extra parameters after the fact. It enters through $W_{PI}(\lambda)$ and $\Delta_{joint}(\lambda)$, or through another comparison theory declared before the target data are interpreted.

That is what “first refusal” means in QCCT. It is not a presumption that the standard theory must win. It is a requirement that a stronger claim cannot be earned while a weaker, physically warranted explanation remains live.

What Can Already Be Tested, and What Remains Open

QCCT does not begin from an empty experimental landscape. Its individual burdens have been approached from several directions, often for reasons unrelated to this proposal. That prior work is useful not because it has already performed QCCT, but because it shows which parts of the experimental architecture are physical capabilities rather than speculative requirements.

The clearest example is sequential qutrit measurement. Jerger and collaborators implemented deterministic binary-outcome measurements on a superconducting transmon qutrit by engineering a readout that distinguished one target state from the complementary two-dimensional subspace while leaving information within that unresolved subspace intact. Their apparatus supported sequential individual measurements and separately quantified measurement incompatibility; the individual

measurement data were also released publicly.²¹ This is close to the physical architecture required by QCCT's secondary coarse-graining test, even though the original experiment targeted contextuality rather than state-update theory.

That secondary test follows a simple property of the generalized-update analysis. If P_x is contained in a coarser projector P_γ , ideal Lüders continuation predicts that inserting the coarse measurement should not alter the later probability of P_x :

$$\Pr(P_\gamma \text{ then } P_x) = \Pr(P_x \text{ direct}).$$

QCCT calls this **coarse-graining invariance**, or CGI, to distinguish the property from broader uses of the word coherence. The Jerger experiment's sequential dichotomic architecture and public individual-outcome data make it a plausible retrospective CGI dataset. But “plausible” matters. The public paper establishes the measurement geometry and the existence of the data; it does not by itself prove that the retained dataset contains every direct and sequential count, ordering field, and selection record required by the QCCT-CGI analysis. Until those raw fields are audited, the honest status is **retrospective CGI candidate**, not completed CGI adjudication.

The same boundary applies to earlier state-update experiments. The NMR experiment discussed above already shows that postmeasurement rules can be experimentally discriminated at a particular device. Its existence removes any novelty claim based merely on testing Lüders updating. What remains distinctive about QCCT-I is the stronger preparation-indistinguishability null: standard linear instruments cannot distinguish G_Z from G_X once those procedures enter the same admitted density operator, whereas the selected nonlinear Gemenge update can.

Process-memory capability is similarly farther along than the proposal requires one to assume. Restricted process-tensor characterization has been demonstrated on superconducting quantum processors, and full multi-time process tomography has

²¹ M. Jerger et al., “Contextuality without nonlocality in a superconducting quantum system,” *Nature Communications* 7, 12930 (2016), DOI 10.1038/ncomms12930. The experiment implemented sequential dichotomic qutrit measurements that projected one target state while preserving information in the complementary two-dimensional subspace. By matching the dispersive responses of the first and second excited states, the apparatus made those readout responses indistinguishable while retaining their mutual coherence. The paper reports deterministic individual outcomes, explicit compatibility tests, and public measurement data at DOI 10.14264/uql.2016.207. QCCT treats those data as a retrospective CGI candidate until their exact retained fields are audited against the QCCT-CGI requirements.

now been demonstrated on superconducting qubits.²² The latter work also provides an open analysis repository containing experimental data and reconstruction code, making it possible for independent groups to inspect or reuse the process-characterization workflow rather than treating it as a black box.²³ Those datasets do not automatically answer the QCCT preparation-indistinguishability question—the experiments were designed for different targets—but they can already close pieces of the feasibility and comparator burden.

The operational-equivalence side has its own experimental precedent. Aloy and collaborators recently reconstructed the behavior of a superconducting qubit in a generalized-probabilistic-theory framework rather than assuming a trusted quantum state space from the outset. Their analysis makes operational indistinguishability and the tomographic-completeness assumption explicit while tracking the contraction of the experimentally realized state space under decoherence.²⁴ This does not supply QCCT's target data, but it demonstrates a relevant route for constructing and auditing the reference relative to which G_Z and G_X are claimed to represent the same input.

Mazurek and collaborators provide a complementary technique. In their noncontextuality experiment, the primary procedures realized in the laboratory did not satisfy the desired operational equivalences exactly. Rather than treating approximate equality as exact, they constructed secondary procedures as probabilistic mixtures of the experimentally available primaries so that the required equivalences held in the admitted operational representation.²⁵ For QCCT, that is an important methodological option if independently prepared Z- and X-basis Gemenge fail the reference-equivalence tolerance before the continuation test begins.

²² G. A. L. White et al., “Demonstration of non-Markovian process characterisation and control on a quantum processor,” *Nature Communications* 11, 6301 (2020), DOI 10.1038/s41467-020-20113-3; C. Giarmatzi, T. Jones, A. Gilchrist, P. Pakkiam, A. Fedorov, and F. Costa, “Multi-time quantum process tomography on a superconducting qubit,” *Quantum* 9, 1952 (2025), DOI 10.22331/q-2025-12-18-1952. White supplies the restricted-process precedent; Giarmatzi et al. supply the later full multi-time superconducting-qubit demonstration.

²³ The public GitHub repository Christina-Giar/NMN-tomo accompanying Giarmatzi et al. contains the analysis notebook and experimental JSON datasets used to reconstruct the reported multi-time process matrices. Its relevance here is reproducibility and comparator reuse rather than direct QCCT-I closure: the original experiment did not target the G_Z/G_X preparation-indistinguishability question.

²⁴ A. Aloy et al., “Theory-independent monitoring of the decoherence of a superconducting qubit with generalized contextuality,” *Nature Communications* 17, 2474 (2026), DOI 10.1038/s41467-026-69030-x.

²⁵ M. D. Mazurek, M. F. Pusey, R. Kunjwal, K. J. Resch, and R. W. Spekkens, “An experimental test of noncontextuality without unphysical idealizations,” *Nature Communications* 7, 11780 (2016), DOI 10.1038/ncomms11780.

Taken together, these experiments change the feasibility posture of QCCT. The proposal does not require a laboratory to invent sequential superconducting measurement, operational-equivalence reconstruction, or multi-time process characterization from scratch. Those capabilities exist in neighboring forms. What remains open is their integration around the specific continuation question.

The minimum proposed experiment, QCCT-I, is correspondingly modest in hardware concept: one qubit, two real randomized preparation Gemenge associated with the same standard density operator, one admitted first measurement, and at least one later probe. The difficult work is not the abstract pulse sequence. It is establishing that the preparation procedures really enter the same declared quantum reference, that the apparatus cannot access their identity through an uncontrolled side channel, and that any temporal memory capable of carrying that identity has been represented strongly enough for the claimed comparator.

The qutrit experiment therefore remains useful without being compulsory. QCCT-CGI becomes a secondary module: it tests the coarse-graining property associated with Lüders continuation and may be susceptible to retrospective analysis using existing sequential qutrit data. It can provide an independent state-update diagnostic even if no laboratory chooses to implement QCCT-I immediately.

A thicker composite-system question should be staged later. Fiorentino and Weigert identify composition compatibility as an additional condition needed when extending their single-system analysis to composite systems.²⁶ Experimentally, however, that step introduces entangled-state admission, subsystem coupling, crosstalk, spectator effects, local-versus-global measurement identity, and a larger tomography burden. None of those complications is necessary to ask the cleaner single-qubit preparation-indistinguishability question. QCCT-II therefore treats composition compatibility as a follow-on burden rather than allowing it to contaminate the minimum experiment.

Several practical limitations remain genuinely open. A particular hardware platform may not be able to erase preparation-side information strongly enough for G_Z and G_X to share the required reference. A full process-tensor intervention basis may be

²⁶ Fiorentino and Weigert, *Physical Review A* 113, 012204 (2026). Their single-system analysis identifies the coarse-graining property used here as CGI; for composite systems, composition compatibility supplies an additional constraint in their uniqueness result for Lüders updating. QCCT therefore retains composition compatibility as a scientifically motivated follow-on module rather than requiring it for the minimum single-qubit experiment.

too expensive, forcing a restricted comparator. Drift may prevent one stationary instrument from describing an entire acquisition window. Conditioning on P_0 discards part of the data and therefore raises the sampling burden. The λ range worth testing depends on achievable readout error, control fidelity, side-channel bounds, and the extent to which other experiments may already constrain the selected foil.

These are not details the proposal should invent in advance. They are experimental burdens to be closed by the specialists who own the relevant platform. A laboratory may find that one of them is presently unclosable. That result would not make the question illegitimate. It would identify the exact physical obstruction preventing the target comparison from being entered on that apparatus.

The same is true of existing data. A published dataset may contain enough information to close one module and not another. Sequential qutrit outcomes may support CGI without containing the Gemenge preparations required for QCCT-I. A multi-time process-tomography dataset may validate memory reconstruction without containing the target preparation pair. An operational-equivalence dataset may validate the reference methodology without containing an intermediate measurement at all. QCCT does not ask those datasets to answer questions they were never designed to answer.

That modularity is not incidental. Once the burdens are separated, an existing dataset may close one, a new experiment another, and a negative feasibility result may identify exactly where a third cannot yet be entered. The scientific unit of progress is therefore a warranted local closure of a declared physical burden. The next section makes explicit how such closures can accumulate without pretending that heterogeneous experiments are one experiment.

An Open-Source Experiment That Does Not Require One Laboratory

The modularity described above is not merely a convenient way to divide experimental labor. It follows from the structure of the problem itself. Calibration, preparation equivalence, instrument characterization, process-memory reconstruction, coarse-graining behavior, and composition compatibility are different physical

burdens. A result that closes one of them at a declared apparatus, level, and scope does not become less scientific because another laboratory must close the next one.

QCCT is therefore designed around **local closure rather than institutional ownership of the whole experiment**. One group may establish that a preparation protocol realizes the required operational equivalence. Another may characterize a measurement instrument. Another may demonstrate that a finite-horizon process tensor accounts for the relevant apparatus memory. An archival-data group may close the CGI module retrospectively. A later experiment may address composition compatibility. None of those contributions has to pretend to answer every other question in order to be useful.

This point matters especially for experimental foundations, where different laboratories often possess different strengths. The apparatus best suited to calibrating an operational reference need not be the apparatus best suited to multi-time process tomography. An older dataset may contain exactly the sequential outcomes required for one QCCT module while lacking the preparation metadata required for another. A theory group may derive a new admissible update family without owning any hardware at all. The open protocol allows each contribution to retain its own evidentiary identity.

That does not mean results can simply be pooled across experiments because they share a label. Closure is local to a physical seat. A calibration result obtained for one device does not automatically apply to another. A process tensor reconstructed over one intervention space does not characterize a larger one. Operational equivalence established at one reference does not silently survive a different preparation procedure, a different calibration epoch, or a wider system boundary. A partial result can serve as an antecedent for another result only if the relevant identities and interfaces are themselves warranted.

For that reason, every QCCT return carries more than a verdict. It must identify the anchor and companion versions under which it was produced, the physical carrier or apparatus, the level and scope of the claim, the operational reference, the provenance of the data and calibration, the assumptions under which the result holds, and the burdens that remain open. A downstream group that wants to inherit the result must show that those conditions remain compatible with its own experiment. If they do

not, the earlier result remains valid at its original seat; what stays open is the interface between the two experiments.

This produces a form of distributed empirical closure without pretending that heterogeneous data are one experiment. A superconducting-qubit laboratory might establish preparation equivalence and the primary QCCT-I null. A separate group might demonstrate a stronger process-memory comparator. A qutrit dataset might close CGI. A later composite-system experiment might address composition compatibility. These results can accumulate into a stronger adjudication architecture, but only through explicit translation of object, scope, reference, and evidence.

The distinction is especially important when results disagree. If two groups return incompatible outcomes under apparently similar conditions, QCCT does not instruct them to average the disagreement away. It opens a new discrimination problem. Are the physical seats actually equivalent? Did the same measurement instrument operate? Were the same side channels controlled? Did one experiment include a degree of freedom omitted by the other? Did the QCCT instructions change between implementations? A genuine residual conflict is itself information about where the current physical description is incomplete.

The decomposition used here follows a simple discrimination rule: a candidate burden counts as distinct only if it performs a distinct job, has a weakest honest physical shape, and loses a specific capacity when it fails while neighboring structure survives. The surviving burdens are then assigned to the narrowest physical or theoretical level capable of answering them. Calibration stays with measurement physics. Born weighting stays with quantum probability. Outcome standing stays with the theory that asserts it. Continuation stays with the measurement instrument or successor process. The method is designed to prevent one job from being silently substituted for another, not to replace the underlying physics.²⁷

The same discipline governs the machine layer released with the proposal. The public machine file contains separate Seam G/Born and QCCT-I contracts and adapters beneath the unchanged Universal Kernel. Their scientific objects remain distinct even though they are packaged in one Lean file. Their generic dispositions remain **NOT**

²⁷ The burden-separation and scientific scope used here are established in this anchor artifact, while the QCCT Audit and Execution Companion supplies the corresponding execution, result-transfer, STOP, and reporting procedures. Their role is methodological: they keep claims local to the evidence and scope that support them. The quantum-mechanical content of QCCT remains supported by the physics and measurement sources cited throughout this paper.

YET ADJUDICABLE because the domain-owned probability requirements and the real QCCT empirical requirements are intentionally left open. A clean machine run establishes that those encoded architectures coexist coherently with the compatible kernel used to express them. It does not derive the Born rule, establish that the laboratory has satisfied the QCCT entry requirements, prove that the λ -family is physically correct, or establish any experimental result.²⁸

That distinction is deliberate. The machine layer audits the encoded architecture of the question. The experiment governs the physical answer.

Public auditability then becomes more than access to source files. A critic need not accept the entire program in order to challenge it. The preparation-indistinguishability null can be attacked independently. The Gemenge treatment can be attacked independently. The process-tensor comparator can be attacked independently. The side-channel burden can be shown unclosable on a particular apparatus. An existing dataset can be shown to contain—or not contain—the fields required by one module. A laboratory can return a negative feasibility result without being asked to endorse the broader framework.

The proposal is therefore intended to fail locally if it fails at all. A defect in CGI should not silently destroy the preparation-indistinguishability experiment. A side-channel problem on one superconducting platform should not establish that the question is untestable in principle. A formula error in the generalized foil should reopen that comparison without invalidating the underlying standard-instrument distinction. Conversely, no successful local result is allowed to inflate into whole-program closure.

That combination—local failure, local closure, and explicit inheritance—is what makes the open-source architecture scientifically useful. Different groups can contribute to the same adjudication problem without first agreeing on an

²⁸ The release package includes SF_Quantum_Measurement_Contracts_and_Adapters_v1_0.lean, appended beneath the unchanged compatible Structural Flow Universal Kernel. The consolidated file contains separate Seam G/Born and QCCT-I machine contracts and adapters and was elaborated cleanly twice before publication. Their encoded generic dispositions remain NOT YET ADJUDICABLE because the relevant domain and empirical requirements are intentionally open. The machine result certifies the encoded machine architecture only; it does not derive the Born rule, certify the λ derivation, establish apparatus feasibility or empirical preparation equivalence, close the process comparator, or establish any experimental outcome.

interpretation of quantum mechanics, adopting the same apparatus, or assigning one laboratory authority over the entire measurement problem.

What a Result Would Mean

The strongest feature of QCCT may be that nearly every scientifically interesting outcome is informative without being promoted into a universal verdict.

The experiment can fail before the target witness is ever reached. If G_Z and G_X cannot be shown to represent the same admitted input at the declared reference, the correct result is not a noisy estimate of W_{PI} . The preparation-equivalence burden remains open. If the two procedures are equal as qubit states but leave distinguishable preparation history in a resonator, higher level, controller, or other admitted external degree of freedom, the correct result is a preparation-side-channel finding. In both cases the experiment has located why the standard null was not yet applicable.

If the entry conditions close and the sequential data are predicted by one fixed standard quantum instrument, QCCT returns standard-instrument closure at that scope. That result would place a bound on any preparation-dependent continuation effect accessible to the experiment. If an ideal Lüders model fails while another ordinary instrument succeeds, the experiment has characterized a non-Lüders measurement device—not discovered new quantum mechanics.

A fixed instrument may fail while a finite-horizon process tensor succeeds. In that case, the experiment has located temporally extended apparatus memory that the simpler model omitted. Again, that is a substantive physical result. It says that the measurement cannot be treated as one stationary outcome-conditioned operation over the tested sequence, while the larger standard quantum process description remains sufficient.

If the registered process comparator fails, the next question is whether the declared reference was physically incomplete. An independently observed resonator memory, leakage level, spectator coupling, or controller degree of freedom may warrant an expanded system boundary. If the expanded reference then closes, the anomaly has been localized rather than erased: the earlier description failed because it left out a physically relevant carrier of preparation history.

Only after those routes have been discharged can a residual earn the statement that preparation indistinguishability is violated **at the tested surface**. Even then, the experiment does not leap directly to a theory of measurement. It has shown that two proper-mixture preparation procedures represented by the same admitted standard quantum state produce different outcome-conditioned continuations after the registered ordinary explanations have failed at that scope.

The named generalized-update family can then be tested against the shape and magnitude of the residual. Because the family predicts

$$W_{PI}(\lambda) = \lambda / (1 + \lambda),$$

a null result can bound λ under the declared foil assumptions, while a nonzero result can be compared with the corresponding λ dependence. Agreement would support that family relative to the alternatives actually entered into the comparison; disagreement would constrain or reject it. Neither outcome makes the family exhaustive.

The strongest possible result contemplated by QCCT is therefore intentionally bounded: **the registered ordinary quantum comparator is insufficient at the declared physical scope**. Such a result would warrant an additional physical answer. It would not, by itself, establish what that answer is.

This distinction matters because several much larger claims are logically downstream of the experiment and remain unearned even in its strongest branch. QCCT does not by itself establish that quantum mechanics is universally false. It does not establish that wave-function collapse is or is not a literal physical process. It does not select Everettian, Bohmian, relational, collapse-model, operational, or other interpretations. It does not exclude every possible nonlinear update theory. It does not prove that every environment has been exhausted. It does not derive the Born rule or settle whether exactly one outcome has theory-wide physical standing.

Those questions were separated at the beginning because they are different questions. Recombining them at the end merely because an experiment produced an interesting residual would undo the central methodological result of the paper.

The converse is equally important. A null result is not scientifically empty because it fails to overturn quantum mechanics. A strong QCCT-I null would place an

experimentally audited constraint on preparation-dependent continuation for a named generalized-update family. Standard-instrument closure would characterize the actual measurement dynamics. Process-tensor closure would establish how far temporal memory can account for the sequence. CGI closure would constrain a different state-update property. A laboratory that closes only one of these burdens has still added a warranted piece to the measurement architecture.

That is why the proposal does not have a single success condition. Its central requirement is that the result mean exactly what the evidence allows it to mean.

Conclusion: The Question Left to the Experiment

Quantum measurement becomes harder to discuss when several physical jobs are allowed to travel under one name. Calibration is not record formation. Record formation is not unique outcome standing. Outcome standing is not probability. One-shot probability is not outcome-conditioned continuation. Once those burdens are separated, disagreement about one of them no longer has to be converted into disagreement about all of them.

QCCT isolates one consequence of that separation that can be asked directly in the laboratory. Two real proper-mixture procedures can be represented by the same density operator in standard quantum mechanics. A fixed standard quantum instrument then has no access to which decomposition produced that state. A published generalized nonlinear update family need not erase the same distinction.

The experiment therefore asks a narrow physical question:

If two physically different proper-mixture preparations are represented by the same admitted quantum state, does nature erase that difference from the statistics of what happens after the same measurement outcome?

Ordinary quantum physics is given first refusal. Preparation mismatch, side channels, non-Lüders instruments, temporal memory, and independently warranted reference expansion all receive their own opportunities to account for the data before a generalized continuation rule earns attention.

No single laboratory has to close every burden. Existing datasets can answer some. New experiments can answer others. A negative feasibility result can identify where

the present QCCT protocol cannot yet be implemented. Each contribution remains local to the seat and scope at which it was earned, and each can become an antecedent for later work only through an explicit interface.

The proposal therefore does not offer a new declaration about what measurement “really is.” It offers a way to stop several different measurement questions from answering for one another, and a public experimental protocol for one of the questions that remains.

The answer belongs to the experiment.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation and revision of this manuscript and its accompanying formal materials, the author used generative AI tools to assist with drafting support, language refinement, structural editing, experimental-design organization, formalization support, machine-audit organization, consistency checking, and clarity. All substantive arguments, scientific claims, experimental-design choices, formalization choices, scope decisions, and final editorial judgments are the author's own. The author reviewed and approved the final manuscript and accompanying formal materials and takes full responsibility for their content.

**Quantum Measurement: Separating the Physical
Burdens and Testing Outcome-Conditioned
Continuation**

Signed,

Rick